MIL-C-10541D
24 August 1978
SUPERSEDING
MIL-C-10541C
9 January 1970

MILITARY SPECIFICATION

COOLING UNIT, STEREOPLOTTER PROJECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cooling unit for use with stereoplotter projectors.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-12044	- Chests, Plywood.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Standard No. 210 - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 30 West University Drive, Arlington Heights, IL 60004.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The stereoplotter projection cooling unit (hereinafter called "cooling unit") shall consist of an electrically driven blower operating in a carrying case, with a built-in air filter and an extension cord.

3.2 First article (first produced cooling unit(s)). The contractor shall furnish one or more cooling units as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce cooling units that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Environmental conditions.

3.4.1 Storage temperatures. The cooling unit shall be capable of prolonged storage at not more than minus 65° F (minus 54° C) and prolonged storage at not less than 155° F (69° C) without incurring difficulty in operation or damage to the unit.

3.5 Blower. The blower shall be of the centrifugal type and shall have a rated capacity, without the filter, of not less than 325 cubic feet per minute (cfm) at 1-inch of mercury (Hg) static pressure.

3.6 Motor. The motor shall be a continuous-duty 115/230-volt, single-phase type, 60-Hertz alternating current, 1/3 horsepower motor conforming to CC-M-1807, Type I, Class 1, Style A, Kind 2, Group C, and Design J.

3.7 Filter. An air filter shall be mounted on the air intake side of the case. The filter shall not reduce the capacity of the blower more than 8 percent from its rated capacity. Two replacement air filters shall be furnished with each unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

3.8 Extension cord. A 15-foot extension cord shall be furnished. The extension cord shall have a 3-wire, polarized, locking type cord connector body on one end. The connector shall fit the motor plug in the case (see 3.9). The other end of the cord shall have a standard 3-wire grounding type plug. All electrical equipment and wiring shall conform to the National Electrical Code, NFPA No. 70.

3.9 Case. A case shall be provided to house the cooling unit, extension cord, and two replacement filters. The case shall permit operation of the unit while encased. The filter, blower, and hose connection shall be integrally connected to prevent air within the case from entering the cooling system. A locking-type, flush-base, male motor plug shall be provided in a recessed panel on the outside of the case for connection to line voltage (see 3.8). A switch shall be provided adjacent to the plug for turning the blower on and off. Provision shall be made in the side of the case, or on the blower, for connecting a 3-1/2-inch inside diameter flexible hose. When the hose connection is on the outside of the case, the opening shall be recessed, and a sliding cover shall be provided to cover the opening. The case shall conform to MIL-C-12044 and the following requirements:

(a) 1/2-inch plywood.
(b) Marking:

U. S.
COOLING UNIT
STEREOPLOTTER

(c) Dimensions, location of external fixtures, and interior compartmentation shall be selected by the contractor to suit the application.

3.10 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference emission characteristics of the cooling unit shall conform to MIL-STD-461.

3.11 Identification marking. The cooling unit shall be identified in accordance with MIL-STD-130.

3.12 Treatment and painting. The portions of the cooling unit normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

3.13 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.14 Workmanship. All parts, components, and assemblies of the cooling unit shall be clean and free from sand, dirt, fins, sprues, scale, flux, rust, and other harmful extraneous material, and cracks. External surfaces shall be free from burrs, sharp edges, and corners, except where sharp edges or corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced cooling unit inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

4.3 First produced cooling unit inspection.

4.3.1 Examination. The first produced cooling unit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced cooling unit shall be tested as specified in 4.5.2.1 and 4.5.2.2 and, when specified herein, 4.5.2.3. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The cooling unit(s) shall be examined for the following defects:

Major

101. Material not as specified.
102. Blower not as specified.
103. Motor not as specified.
104. Filter not as specified.
105. Extension cord not as specified.
106. Case not as specified.
107. Assembly not complete.
108. Parts not interchangeable.
109. Workmanship not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Storage temperature. Place the carrying case and cooling unit in a cold test chamber. The chamber shall be at an initial temperature of from 65° to 75° F (18° to 24° C). Lower the temperature of the chamber to minus 65° F (minus 54° C) and maintain this temperature for not less than 12 hours. At the end of this period, remove the carrying case and cooling unit and allow them to return to the initial temperature. Maintain initial temperature for not less than 4 hours. Place cooling unit and carrying case in a test chamber and raise the temperature to 155° F (69° C). Maintain this temperature for not less than 4 hours. Remove the cooling unit and carrying case from the chamber into an ambient temperature equal to the initial temperature. Examine and operate the cooling unit and examine the carrying case. Any damage to the cooling unit or case or difficulty in operation of the cooling unit shall constitute failure of this test.

4.5.2.2 Capacity. Test the blower for rated capacity in accordance with AMCA Standard No. 210. A rated capacity of less than 325 cfm at 1-inch of Hg static pressure shall constitute failure. A reduction of more than 8 percent of rated capacity of the blower when the cooling unit, with filter, is operated shall constitute failure of this test.

4.5.2.3 Electromagnetic interference. When specified herein, the first produced cooling unit(s) shall be tested to determine conformance to 3.10. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Components not contained within the case.
111.	Box not as specified for Level A or Level B.
112.	Strapping not as specified for Level A.
113.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Each cooling unit, with extension cord and the two replacement filters contained in the case, and the case lid secured, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.1.2 Level B. Each cooling unit, with extension cord and the two replacement filters contained in the case, and the case lid secured, shall be packed in a close-fitting box conforming to PPP-B-640, Class 2, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.1.3 Commercial. The cooling unit shall be packed in accordance with MIL-STD-1188.

5.2 Marking.

5.2.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The cooling units are intended for cooling the lamp housings of multiplex projectors which are used in the compilation of topographic maps.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced cooling unit and number of units required (see 3.2).
(c) When control of electromagnetic interference is required (see 3.10).
(d) Color required (see 3.12).
(e) Degree of packing required (see 5.1).
(f) When Finish A strapping is required (see 5.1.1).

6.3 First produced cooling unit(s). Any changes or deviations of production cooling units from the approved first produced cooling unit(s) during production will be subject to the approval of the contracting officer. Approval of the first produced cooling unit(s) will not relieve the contractor of his obligation to furnish cooling units conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each cooling unit.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of cooling units.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA - GS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 6675-0476

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-10541D Cooling Unit, Stereoplotter Projection

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:12Z
Check the source to verify that this is the current version before use.